

Q6.

```
→ public ListNode rotateRight(ListNode head, int k) {  
    if (head == null || head.next == null, k == 0) {  
        return head;  
    }
```

```
    int length = 1;  
    ListNode tail = head;  
    while (tail.next != null) {  
        tail = tail.next;  
        length++;  
    }
```

```
    tail.next = head;  
    k = k % length;  
    if (k == 0) {  
        tail.next = null;  
        return head;  
    }
```

```
    ListNode newTail = head;  
    for (int i = 0; i < length - k - 1; i++) {  
        newTail = newTail.next;  
    }
```

```
    ListNode newNode = newTail.next;  
    newTail.next = null;  
    return newNode;  
}
```

$K = 2$

head $\rightarrow$ ① $\rightarrow$ ② $\rightarrow$ ③ $\rightarrow$ ④ $\rightarrow$ ⑤

Base case:-

Step 1 $\rightarrow$ Agle head == null hai to usme kuch changes nhi hoga jaisa hai paira ki setion kar do.

Step 2:- head ka length find karo aur uska last node. i.e length find karne ke liye ek ~~length = 0~~ ~~length = 0~~ karenge aur last node find karne ke liye ek node create karenge aur head ke equal kar denge uske baad loop use karenge jab tak (while (temp.next != null)) aur temp = temp.next se update kar denge aur length ++ kar denge.
i.e.

length = 1

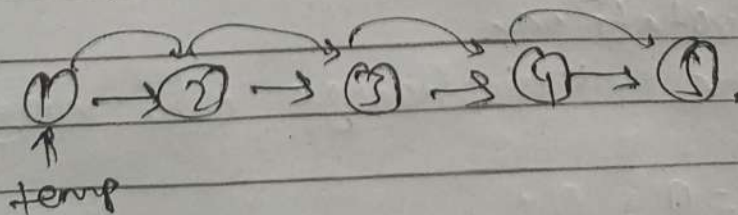
ListNode temp = head;

while (temp.next != null) {

temp = temp.next;

length++

}

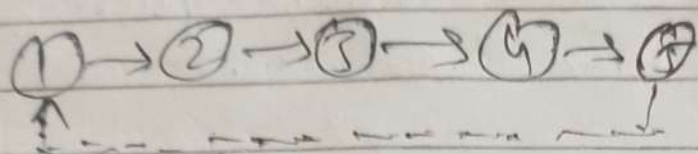


length = 5

temp = 5 (last node)

Step 3:- make a circle, unke liye hm

temp.next = head;



Step 4:- Reduce K. (avoid unwanted full rotation)

$K = K \% \text{length};$

→ this gives remainder,
i.e. $K = 2 \% 5 = 2.$

Yehi pe divide possible hi nhi hai.
→ So, 2 milega as a remainder.

if (

let $K = 10$ & ~~len~~ length = 5.

$K = K \% \text{length}$

$= 10 \% 5 = 0$ (remainder.)

- matches full rotation
- ~~list~~ list me koi change nhi hoga.

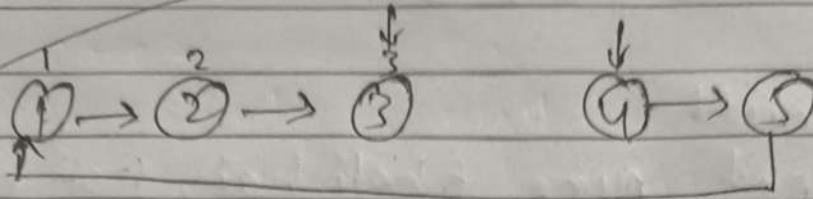
if

if ($K == 0$)

temp.next = null;
return head;

}

Step 5:- Find new tail.
null. new temp = head.



Phir ek loop lagayenge.

for (int i = L; i < length - K; i++)

↳ $i < 5 - 2$

$i < 3$.

i jo hai wo 1 se suru hoga or $i < 3$ tak jayega. aur jise node pe chuke ga usko new tail point kar denge or new temp.

Step 6:- New head. ke liye.

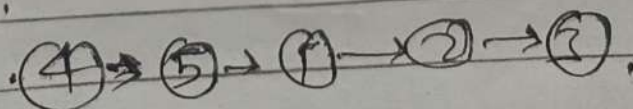
ListNode newNode = new temp.next;

To new temp = 3 hai

To new head = 4 lagayega.

Step 7:- Break the circle :-
new temp.next = null.

So finally.



Edge case:-

Case 1:- head = null, K = 3.

Output = null

kyunki head pura bhalai hai, koi element ya node nhi hai Isliye rotation bhi nhi hoga.

Case 2:- Single node.

head = 1, K = 100.

Time ek hi node hai toh rotation se koi farak nhi padega.

Case 3:- head = 1 → 2 → 3, K = 0.

Rotation 0 hai, toh jo head hai wohi return hoga.

Case 4:- head = 1 → 2 → 3 → 4 → 5, K = 5.

Here, length = 5
K = 5

$$K = K \% \text{length}$$

$$= 5 \% 5$$

$$= 0$$

Same as case 3.